ELEC 3125 - Signals & Systems

Lecture 10 - Part 1

Continuous-Time Fourier Transform

Periodic signal => represented as linear combination of complex exponentials (harmonically related)

Aperiodic signal => represented as linear combination of complex exponentials => infinitesimally close in frequency.

=> Σ => $\int$

As period becomes infinite => Fourier Series becomes an integral.

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega \qquad \text{(Inverse Fourier transform)}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \qquad \text{(Fourier transform)}$$

Examples of Continuous-Time Fourier Transform

Consider: $x(t) = e^{-at}u(t)$, $a>0$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} e^{-at}u(t)\,e^{-j\omega t}dt = \int_0^{\infty} e^{-at}(1)e^{-j\omega t}dt = \int_0^{\infty} e^{-(a+j\omega)t}dt, \quad [-(a+j\omega)t]' = -(a+j\omega)$$

$$\Rightarrow 1 = \frac{1}{-(a+j\omega)}[-(a+j\omega)t]'$$

$$\Rightarrow X(j\omega) = \int_0^{\infty} -\frac{1}{(a+j\omega)}[-(a+j\omega)t]'e^{-(a+j\omega)t}dt = -\frac{1}{(a+j\omega)}\int_0^{\infty}[-(a+j\omega)t]'e^{-(a+j\omega)t}dt = \frac{1}{(a+j\omega)}\int_{\infty}^{0}[-(a+j\omega)t]'e^{-(a+j\omega)t}dt$$

$$= \frac{1}{a+j\omega}\left[e^{-(a+j\omega)t}\right]_{\infty}^{0} = \frac{1}{a+j\omega}\left[e^{-(a+j\omega)(0)} - \frac{1}{e^{(a+j\omega)(\infty)}}\right] = \frac{1}{a+j\omega}, \quad a>0$$

$$\Rightarrow \boxed{e^{-at}u(t) \overset{FT}{\longleftrightarrow} \frac{1}{a+j\omega}, \quad a>0}$$

$$X(j\omega) = \frac{1}{a+j\omega}, \quad a>0$$

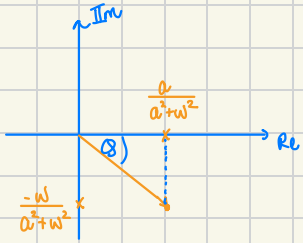
$$|X(j\omega)| = \left|\frac{1}{a+j\omega}\right| = \left|\frac{(a-j\omega)}{(a+j\omega)(a-j\omega)}\right| = \left|\frac{a-j\omega}{a^2+\omega^2}\right| = \left|\frac{a}{a^2+\omega^2} + j\left(\frac{-\omega}{a^2+\omega^2}\right)\right| = \sqrt{\left(\frac{a}{a^2+\omega^2}\right)^2 + \left(\frac{-\omega}{a^2+\omega^2}\right)^2} = \sqrt{\frac{a^2}{(a^2+\omega^2)^2} + \frac{\omega^2}{(a^2+\omega^2)^2}}$$

$$= \sqrt{\frac{(a^2+\omega^2)}{(a^2+\omega^2)^2}} = \frac{1}{\sqrt{a^2+\omega^2}}$$

$$\Rightarrow \boxed{|X(j\omega)| = \frac{1}{\sqrt{a^2+\omega^2}}}$$

$$X(j\omega) = \frac{1}{a+j\omega} = \frac{a}{a^2+\omega^2} + j\left(\frac{-\omega}{a^2+\omega^2}\right)$$

$$\Rightarrow \boxed{\angle X(j\omega) = -\arctan\left(\frac{\omega}{a}\right)}$$

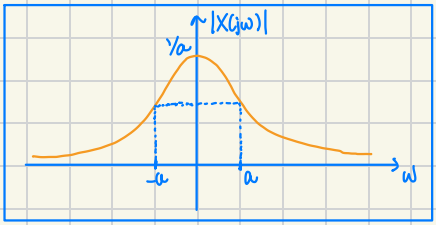


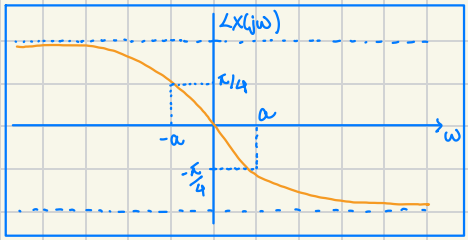
$$\angle X(j\omega) = \theta$$

$$\tan(\theta) = \frac{-\omega}{(a^2+\omega^2)}\,\frac{(a^2+\omega^2)}{a} = -\frac{\omega}{a}$$

$$\Rightarrow \tan(\theta) = -\frac{\omega}{a}$$

$$\Rightarrow \theta = \arctan\left(-\frac{\omega}{a}\right) = -\arctan\left(\frac{\omega}{a}\right)$$





Consider: $x(t) = e^{-a|t|}$, $a>0$

$$X(jw) = \int_{-\infty}^{\infty} x(t) e^{-jwt}\, dt$$

$$|t| = \begin{cases} -t, & t<0 \\ t, & t>0 \end{cases} \Rightarrow x(t) = e^{-a|t|} = \begin{cases} e^{-a(-t)} = e^{at}, & t<0 \\ e^{-at}, & t>0 \end{cases}$$

$$X(jw) = \int_{-\infty}^{\infty} x(t) e^{-jwt}\, dt = \int_{-\infty}^{\infty} e^{-a|t|} e^{-jwt}\, dt = \underbrace{\int_{-\infty}^{0} e^{at} e^{-jwt}\, dt}_{=A} + \underbrace{\int_{0}^{\infty} e^{-at} e^{-jwt}\, dt}_{=B}$$

$$A = \int_{-\infty}^{0} e^{at} e^{-jwt}\, dt = \int_{-\infty}^{0} e^{(a-jw)t}\, dt, \quad [(a-jw)t]' = (a-jw) \Rightarrow 1 = \frac{1}{(a-jw)}[(a-jw)t]'$$

$$= \int_{-\infty}^{0} \frac{1}{(a-jw)}[(a-jw)t]' e^{(a-jw)t}\, dt = \frac{1}{(a-jw)} \int_{-\infty}^{0} [(a-jw)t]' e^{(a-jw)t}\, dt = \frac{1}{a-jw}\left[e^{(a-jw)t}\right]_{-\infty}^{0}$$

$$= \frac{1}{a-jw}\left[\underbrace{e^{(a-jw)(0)}}_{=1} - \cancel{\frac{1}{e^{(a-jw)(\infty)}}}^{\,0}\right] = \frac{1}{a-jw} \Rightarrow \boxed{A = \frac{1}{a-jw}}$$

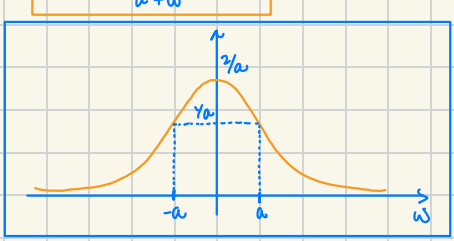
$$B = \int_{0}^{\infty} e^{-at} e^{-jwt}\, dt = \int_{0}^{\infty} e^{-(a+jw)t}\, dt, \quad [-(a+jw)t]' = -(a+jw) \Rightarrow 1 = \frac{1}{-(a+jw)}[-(a+jw)t]'$$

$$= \int_{0}^{\infty} \frac{1}{-(a+jw)}[-(a+jw)t]' e^{-(a+jw)t}\, dt = \frac{1}{-(a+jw)} \int_{0}^{\infty} [-(a+jw)t]' e^{-(a+jw)t}\, dt = \frac{1}{a+jw} \int_{\infty}^{0} [-(a+jw)t]' e^{-(a+jw)t}\, dt$$

$$= \frac{1}{a+jw}\left[e^{-(a+jw)t}\right]_{\infty}^{0} = \frac{1}{a+jw}\left[\underbrace{e^{-(a+jw)(0)}}_{=1} - \cancel{\frac{1}{e^{(a+jw)(\infty)}}}^{\,0}\right] = \frac{1}{a+jw} \Rightarrow \boxed{B = \frac{1}{a+jw}}$$

$$X(jw) = A + B = \frac{1}{(a-jw)} + \frac{1}{(a+jw)} = \frac{(a+jw)}{(a-jw)(a+jw)} + \frac{(a-jw)}{(a+jw)(a-jw)} = \frac{a+\cancel{jw} + a-\cancel{jw}}{(a-jw)(a+jw)} = \frac{2a}{a^2+w^2}, \quad a>0$$

$$\Rightarrow \boxed{e^{-a|t|} \overset{FT}{\leftrightarrow} \frac{2a}{a^2+w^2}, \quad a>0}$$



Consider the signal whose fourier transform is:

$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-W}^{W}(1)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-W}^{W} e^{jt\omega}d\omega \quad . \quad [jt\omega]' = jt \Rightarrow 1 = \frac{1}{jt}[jt\omega]'$$

$$= \frac{1}{2\pi}\int_{-W}^{W}\frac{1}{jt}[jt\omega]'e^{jt\omega}d\omega = \frac{1}{2j\pi t}\int_{-W}^{W}[jt\omega]'e^{jt\omega}d\omega = \frac{1}{2j\pi t}\left[e^{jt\omega}\right]_{-W}^{W} = \frac{1}{2j\pi t}\left[e^{jtW} - e^{-jtW}\right] = \frac{1}{\pi t}\frac{(e^{jtW} - e^{-jtW})}{2j}$$

$$= \frac{\sin(Wt)}{\pi t}$$

$$\Rightarrow \boxed{\frac{\sin(Wt)}{\pi t} \Leftrightarrow X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}}$$

Example: $x(t) = \delta(t)$

$$X(j\omega) = \int_{-\infty}^{\infty}\delta(t)e^{-j\omega t}dt = 1 \Rightarrow \boxed{\delta(t) \Leftrightarrow 1}$$

Example: $x(t) = \delta(t-t_0)$

$$X(j\omega) = \int_{-\infty}^{\infty}\delta(t-t_0)e^{-j\omega t}dt = \int_{-\infty}^{\infty}\delta(t-t_0)e^{-j\omega t_0}dt = e^{-j\omega t_0}\int_{-\infty}^{\infty}\delta(t-t_0)dt = e^{-j\omega t_0}$$

$$\Rightarrow \boxed{X(j\omega) \Leftrightarrow e^{-j\omega t_0}}$$

Prove that: $1 \Leftrightarrow 2\pi\delta(\omega)$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}X(j\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}2\pi\delta(\omega)e^{j\omega t}d\omega = \frac{2\pi}{2\pi}\int_{-\infty}^{\infty}\delta(\omega)e^{j\omega t}d\omega = \int_{-\infty}^{\infty}\delta(\omega)e^{j(0)t}d\omega = \int_{-\infty}^{\infty}\delta(\omega)d\omega = 1 \Rightarrow \boxed{1 \Leftrightarrow 2\pi\delta(\omega)}$$

Prove that: $e^{j\omega_0 t} \Leftrightarrow 2\pi\delta(\omega-\omega_0)$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}X(j\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}2\pi\delta(\omega-\omega_0)e^{j\omega t}d\omega = \frac{2\pi}{2\pi}\int_{-\infty}^{\infty}\delta(\omega-\omega_0)e^{j\omega t}d\omega = e^{j\omega_0 t}\int_{-\infty}^{\infty}\delta(\omega-\omega_0)d\omega = e^{j\omega_0 t}$$

$\delta(\omega-\omega_0) = 1$ when: $\omega = \omega_0$

$$\Rightarrow \boxed{e^{j\omega_0 t} \Leftrightarrow 2\pi\delta(\omega-\omega_0)}$$

The Fourier Transform for Periodic Signals

$$x(t) = \sum_{k=-\infty}^{\infty}a_k e^{jk\omega_0 t}$$

$$X(j\omega) = \sum_{k=-\infty}^{\infty}2\pi a_k\delta(\omega-k\omega_0)$$

$$\Rightarrow \boxed{\sum_{k=-\infty}^{\infty}a_k e^{jk\omega_0 t} \Leftrightarrow \sum_{k=-\infty}^{\infty}2\pi a_k\delta(\omega-k\omega_0)}$$

$$a_k e^{j(k\omega_0)t} \Leftrightarrow a_k 2\pi\delta(\omega-k\omega_0)$$

$$a_k e^{j(k\omega_0)t} \Leftrightarrow 2\pi a_k\delta(\omega-k\omega_0)$$

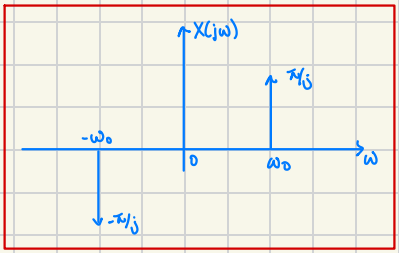
$$\boxed{\sum_{k=-\infty}^{\infty}a_k e^{j(k\omega_0)t} \Leftrightarrow \sum_{k=-\infty}^{\infty}2\pi a_k\delta(\omega-k\omega_0)}$$

Example: Fourier transforms for $x(t) = \sin(\omega_0 t)$

$$x(t) = \sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$$

$$\Rightarrow X(j\omega) = \frac{2\pi\delta(\omega-\omega_0) - 2\pi\delta(\omega+\omega_0)}{2j} = \frac{\pi}{j}\delta(\omega-\omega_0) - \frac{\pi}{j}\delta(\omega+\omega_0)$$

$$\Rightarrow \boxed{\sin(\omega_0 t) \Leftrightarrow \frac{\pi}{j}\delta(\omega-\omega_0) - \frac{\pi}{j}\delta(\omega+\omega_0)}$$

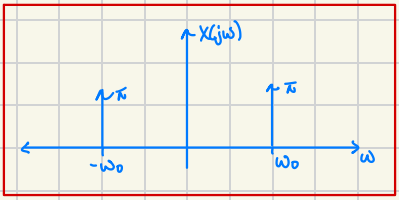


Example: Fourier transforms for $x(t) = \cos(\omega_0 t)$

$$x(t) = \cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

$$\Rightarrow X(j\omega) = \frac{2\pi\delta(\omega-\omega_0) + 2\pi\delta(\omega+\omega_0)}{2} = \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$$

$$\Rightarrow \boxed{\cos(\omega_0 t) \Leftrightarrow \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)}$$



FOURIER TRANSFORM PAIRS

$e^{-at}u(t) \Leftrightarrow \frac{1}{a+j\omega}, a>0$	$e^{-a\lvert t\rvert} \Leftrightarrow \frac{2a}{a^2+\omega^2}, a>0$
$x(t) = \begin{cases} 1, & \lvert t\rvert < T_1 \\ 0, & \lvert t\rvert > T_1 \end{cases} \Leftrightarrow \frac{2\sin(\omega T_1)}{\omega}$	$\frac{\sin(Wt)}{\pi t} \Leftrightarrow X(j\omega) = \begin{cases} 1, & \lvert\omega\rvert < W \\ 0, & \lvert\omega\rvert > W \end{cases}$
$\delta(t) \Leftrightarrow 1$	$\delta(t-t_0) \Leftrightarrow e^{-j\omega t_0}$
$1 \Leftrightarrow 2\pi\delta(\omega)$	$e^{j\omega_0 t} \Leftrightarrow 2\pi\delta(\omega-\omega_0)$
$\sin(\omega_0 t) \Leftrightarrow \frac{\pi}{j}\delta(\omega-\omega_0) - \frac{\pi}{j}\delta(\omega+\omega_0)$	$\cos(\omega_0 t) \Leftrightarrow \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$
$\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \Leftrightarrow \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega-k\omega_0)$	$\sum_{k=-\infty}^{\infty} \delta(t-kT) \Leftrightarrow \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} \delta(\omega-\frac{2\pi k}{T})$